Power and resource tradeoffs

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Power and resource tradeoffs before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Pilot experiments

Without opening the old lesson, explain one key idea from Pilot experiments. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Privacy and data governance

Without opening the old lesson, explain one key idea from Privacy and data governance. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Sensitivity analysis

Without opening the old lesson, explain one key idea from Sensitivity analysis. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Power and resource tradeoffs trace and record the first step you can explain without help.

Use W2: Trace the worked case for **Power** and resource tradeoffs. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Power and resource tradeoffs explanation against the supplied facts before discussing it.

Use the worked case for **Power** and resource tradeoffs. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

One hundred alternative trials detect70 at n30 and85 at n60; unit cost2. Compare.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Power and resource tradeoffs facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

W2 Trace the worked case

> Trace the worked case. Apply the supplied Power and resource tradeoffs facts and keep your reasoning.

For a fixed alternative100 simulations detect62 times at n20 and83 at n40. Per-unit cost3 credits. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Audit the limit

> Audit the limit. Apply the supplied Power and resource tradeoffs facts and keep your reasoning.

A report omits this qualification: **Power** estimates do not repair biased sampling or establish the probability a particular hypothesis is true.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Power and resource tradeoffs facts and keep your reasoning.

One hundred alternative trials detect70 at n30 and85 at n60; unit cost2. Compare.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Power and resource tradeoffs facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Power and resource tradeoffs to Reproducible AI Research Study, identify a limit and choose the next useful test.

Add a labelled **power** and resource tradeoffs **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **power** and resource tradeoffs.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Power and resource tradeoffs

> A simulated detection-and-resource trade-off. A fictional simulation repeats a study under the same assumed alternative effect, using two independent-unit sample sizes. Explain the labelled step before checking the changed condition.

A simulated detection-and-resource trade-off. A fictional simulation repeats a study under the same assumed alternative effect, using two independent-unit sample sizes.

There are 100 **simulation trial**s per design. At n=20, 60 detect the specified effect; at n=40, 80 detect it. Each independent unit costs 2 resource units. Same test and effect assumptions are used.

AI ACADEMY / AI-7/23

Stipulated simulation and resource records

	Independent units	Detections	Trials	Cost units
Smaller design	20	60	100	40
Larger design	40	80	100	80

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/23

Power and resource tradeoffs

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Estimate simulated detection

Observed detection fractions are 60/100=60% and 80/100=80%.



2. Calculate design cost

n=20 costs 40 resource units; n=40 costs 80 units.



3. Compare the trade-off

Doubling cost adds 20 percentage points of simulated detection under these assumptions.

CHANGE ONE THING

Change the assumed effect or variability but keep the sample sizes.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Calculate both simulated detection fractions and resource costs. What additional detection is observed for the extra forty units, and does that establish **power** for an actual classroom study?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Reproducible AI Research Study record.

Project: Reproducible AI Research Study

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A simulated detection-and-resource trade-off. A fictional simulation repeats a study under the same assumed alternative effect, using two independent-unit sample sizes.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Doubling cost adds 20 percentage points of simulated detection under these assumptions.

1. Compare the trade-off

2. Estimate simulated detection

3. Calculate design cost

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Compare the trade-off Doubling cost adds 20 percentage points of simulated detection under these assumptions.

Power is the probability of detecting a specified effect under a specified model and decision rule.

Repeated simulation can estimate it; the estimate depends on effect size, variability, design and test assumptions, not sample size alone.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change the assumed effect or variability but keep the sample sizes. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. The detection fractions need new simulation; the resource costs stay 40 and 80.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The detection fractions need new simulation; the resource costs stay 40 and 80.

Power depends on the data-generating assumptions, while this stated unit-cost calculation does not.

Repair a missing building block

Review Pilot experiments (ai-7/22). Goal: Explain the **mechanism** and terminology of pilot experiments. Short practice: Calculate reviewer agreement on the four pilot answers.

What can the disagreement improve, and why are those revised practice answers not untouched final **evidence**?. Return to this week and explain what you changed.

Review Privacy and data governance (ai-7/20). Goal: Explain the **mechanism** and terminology of privacy and data governance. Short practice: Point to the analysis table and the restricted identity key.

Why are the codes pseudonymous rather than guaranteed anonymous, and what changes if the key is published?. Return to this week and explain what you changed.

Review Sensitivity analysis (ai-7/17). Goal: Explain the **mechanism** and terminology of sensitivity analysis. Short practice: Classify 0.4, 0.6 and 0.8 at cutoffs 0.5, 0.7 and exactly 0.6.

Which equality rule decides the boundary, and can these unlabelled scores identify the best cutoff?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

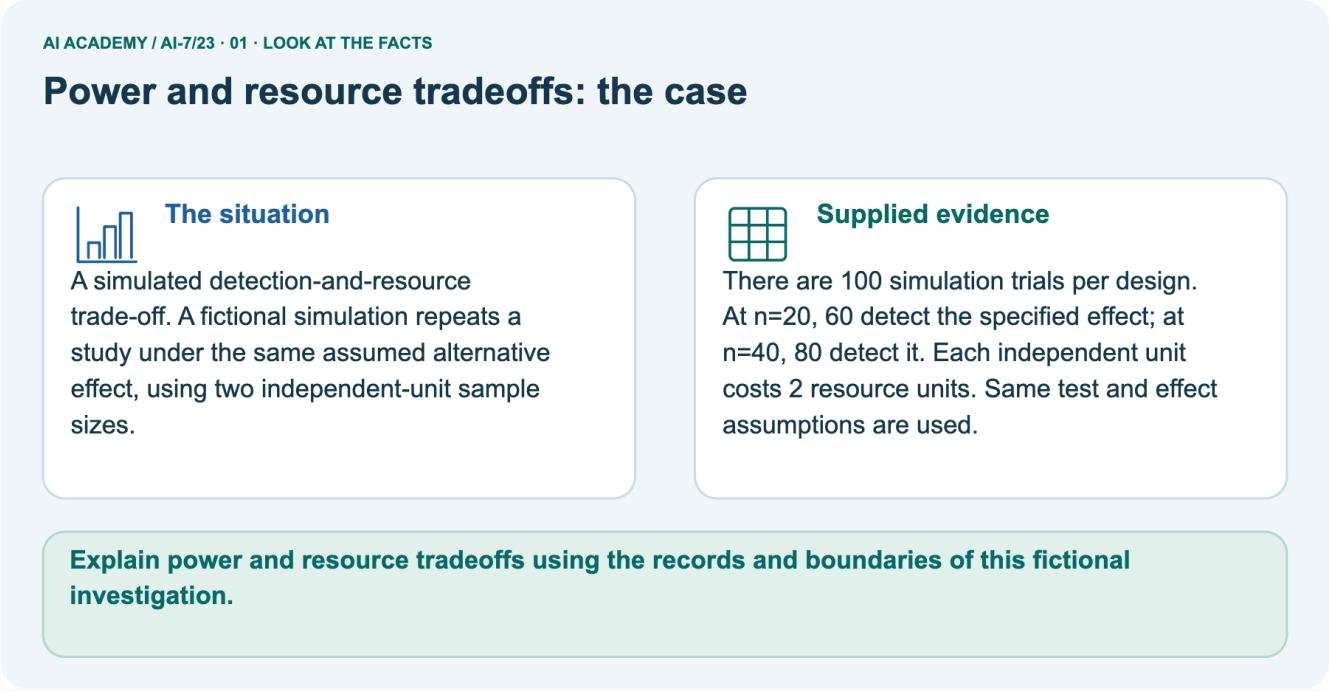
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Power and resource tradeoffs picture from its labels, then record a relevant limit.

01. Power and resource tradeoffs: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

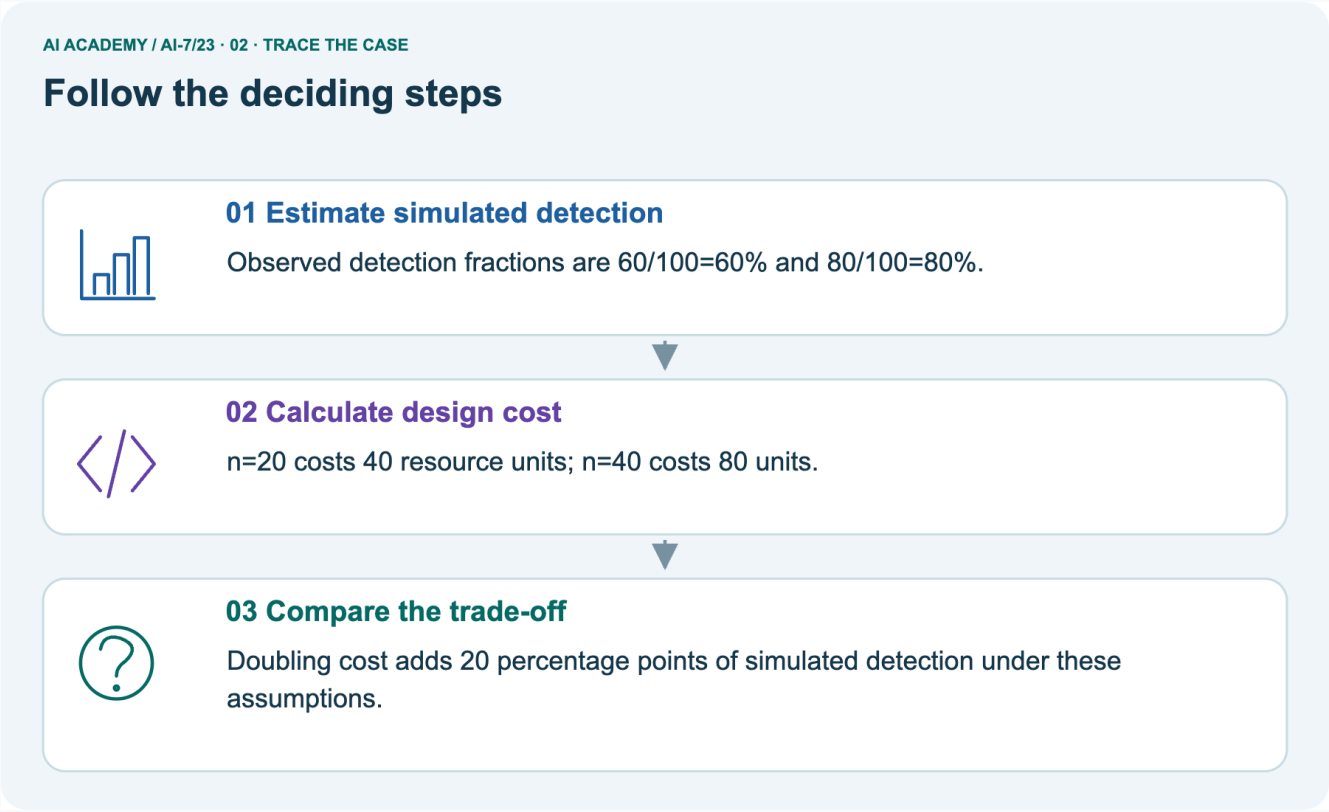
A simulated detection-and-resource trade-off. A fictional simulation repeats a study under the same assumed alternative effect, using two independent-unit sample sizes.

Supplied facts: There are 100 **simulation trial**s per design. At n=20, 60 detect the specified effect; at n=40, 80 detect it. Each independent unit costs 2 resource units. Same test and effect assumptions are used.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



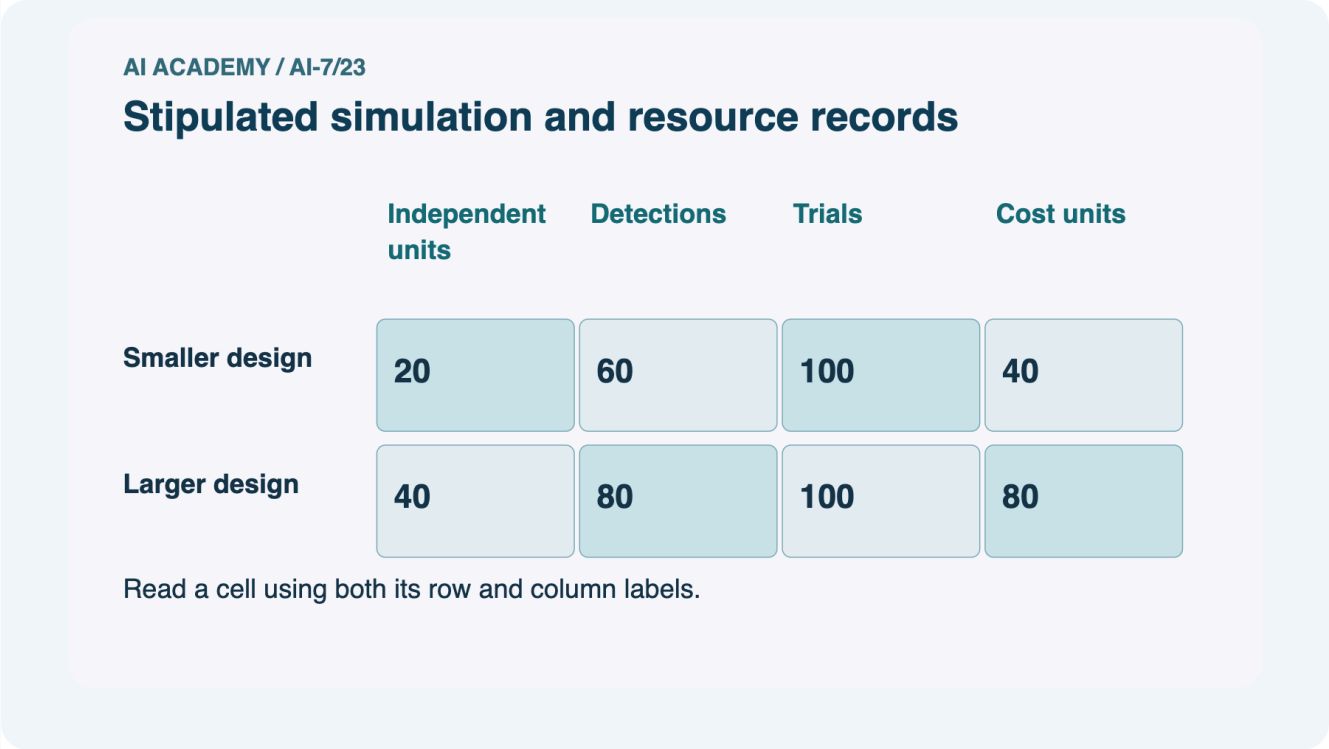
Follow the labelled picture with the matching explanation. Open the vector diagram

Estimate simulated detection: Observed detection fractions are 60/100=**60%** and 80/100=**80%**. → Calculate design cost: n=20 costs 40 resource units; n=40 costs 80 units. → Compare the trade-off: Doubling cost adds 20 percentage points of simulated detection under these assumptions.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Stipulated simulation and resource records



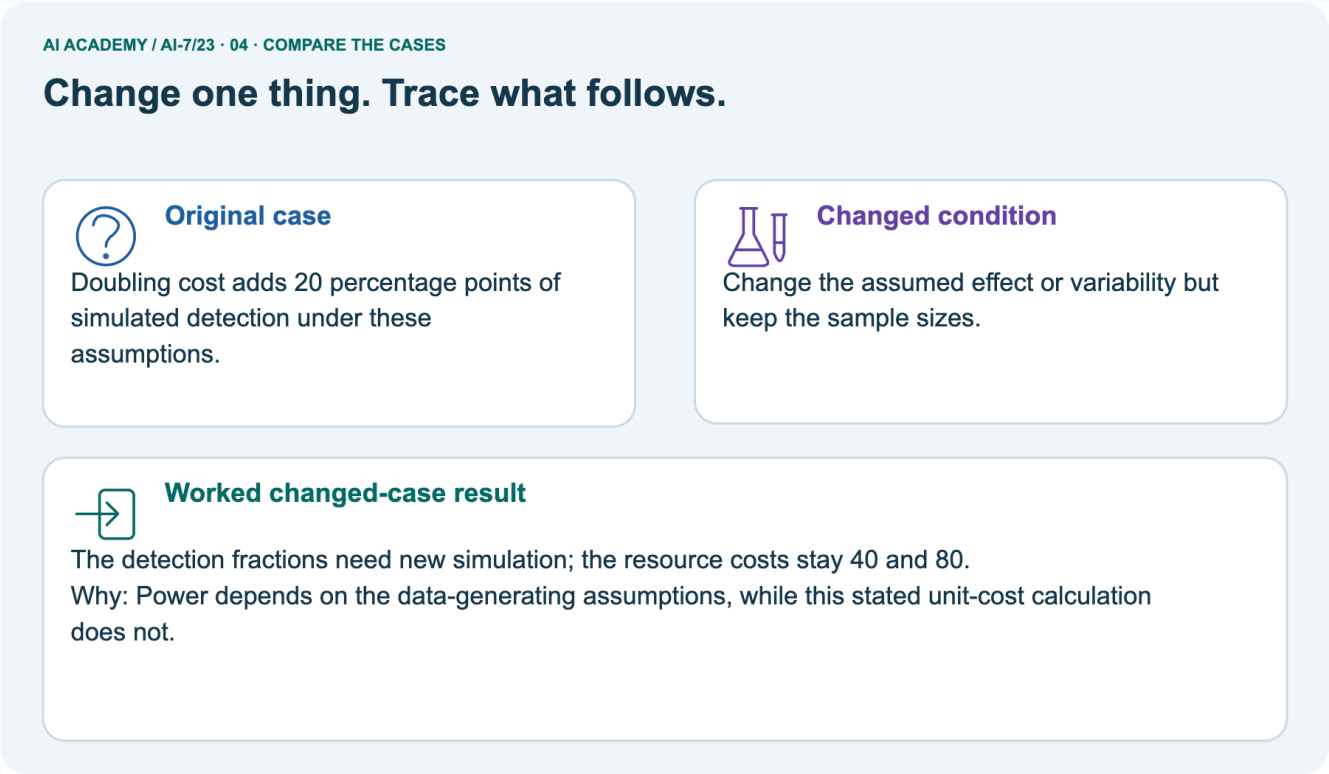
Follow the labelled picture with the matching explanation. Open the vector diagram

Stipulated simulation and resource records. Smaller design: Independent units = 20, Detections = 60, Trials = 100, Cost units = 40. Larger design: Independent units = 40, Detections = 80, Trials = 100, Cost units = 80.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

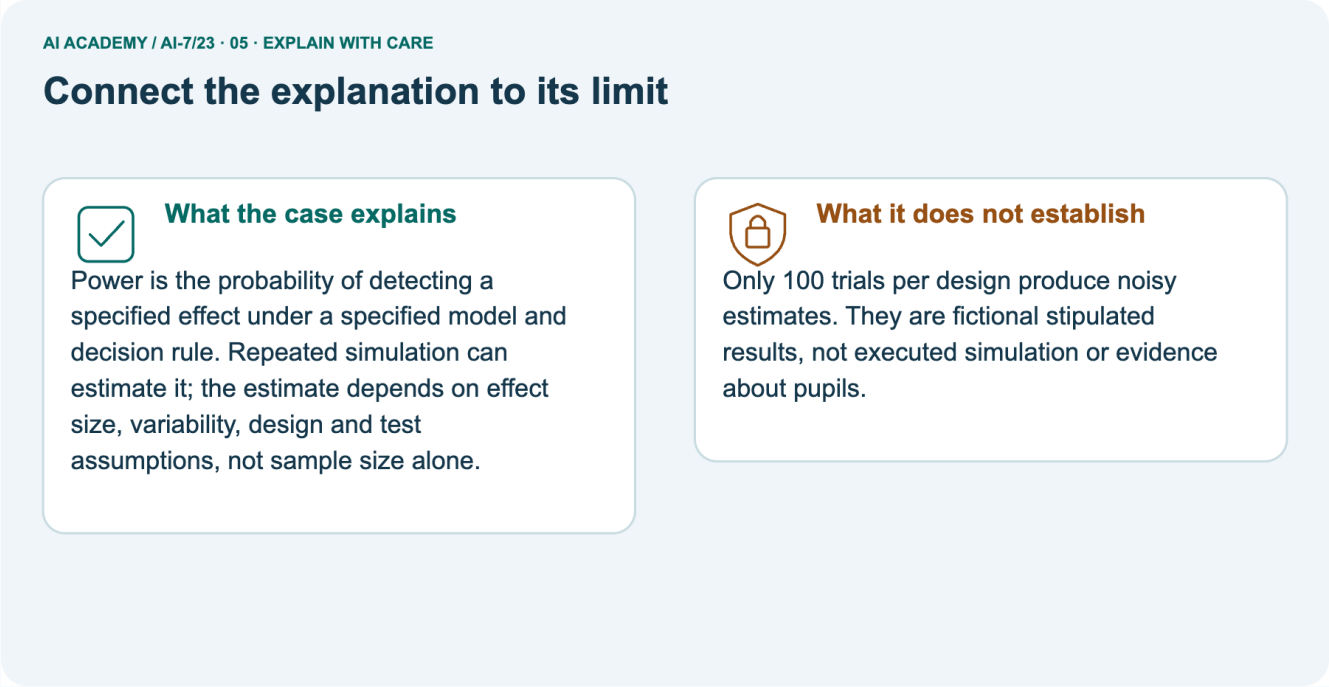
Original case: Doubling cost adds 20 percentage points of simulated detection under these assumptions. Changed condition: Change the assumed effect or variability but keep the sample sizes.

Expected result: The detection fractions need new simulation; the resource costs stay 40 and 80. Reason: **Power** depends on the data-generating assumptions, while this stated unit-cost calculation does not.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Power is the probability of detecting a specified effect under a specified model and decision rule.

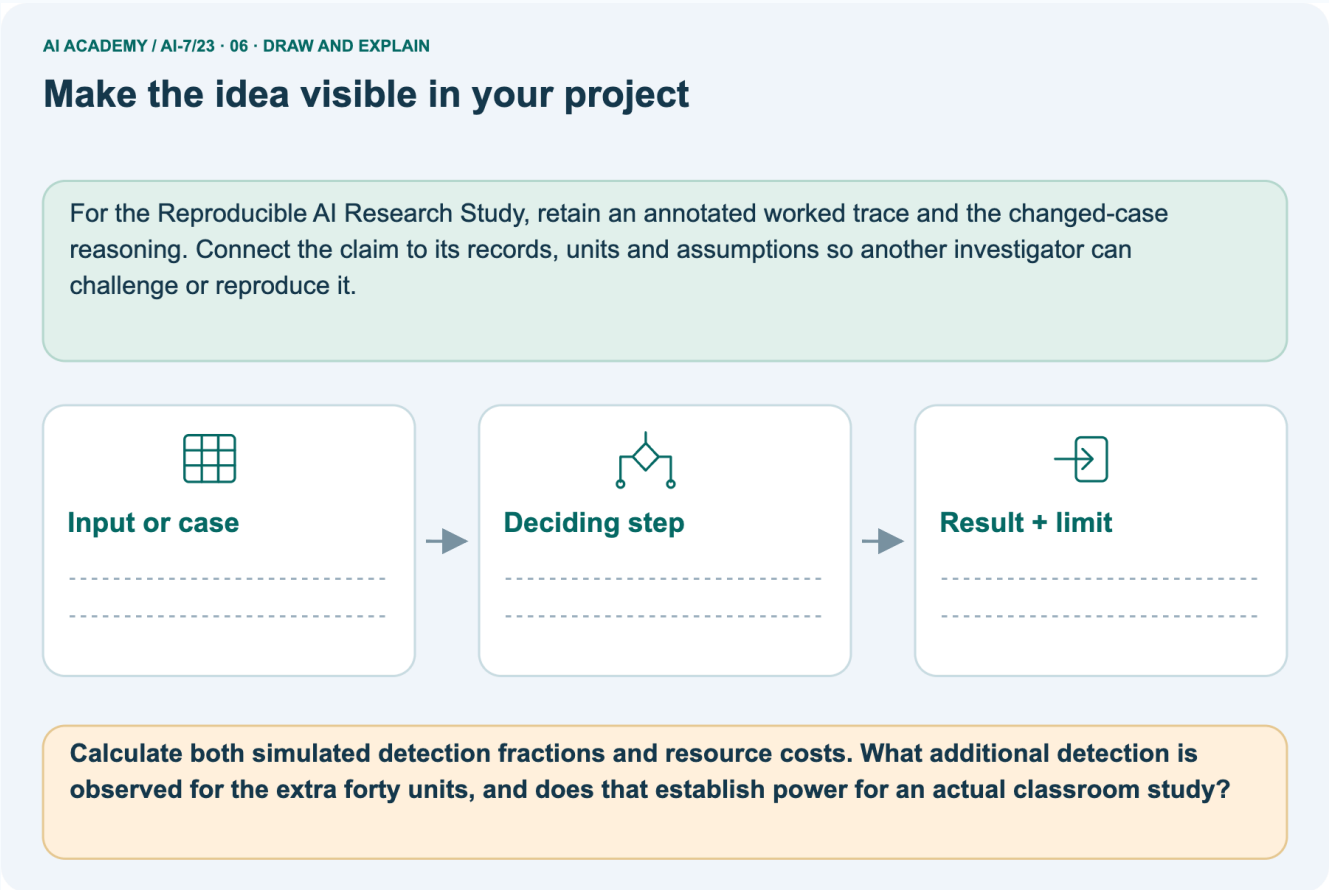
Repeated simulation can estimate it; the estimate depends on effect size, variability, design and test assumptions, not sample size alone. Limit: Only 100 trials per design produce noisy estimates.

They are fictional stipulated results, not executed simulation or **evidence** about pupils.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate both simulated detection fractions and resource costs.

What additional detection is observed for the extra forty units, and does that establish **power** for an actual classroom study?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.